

Payment Management System

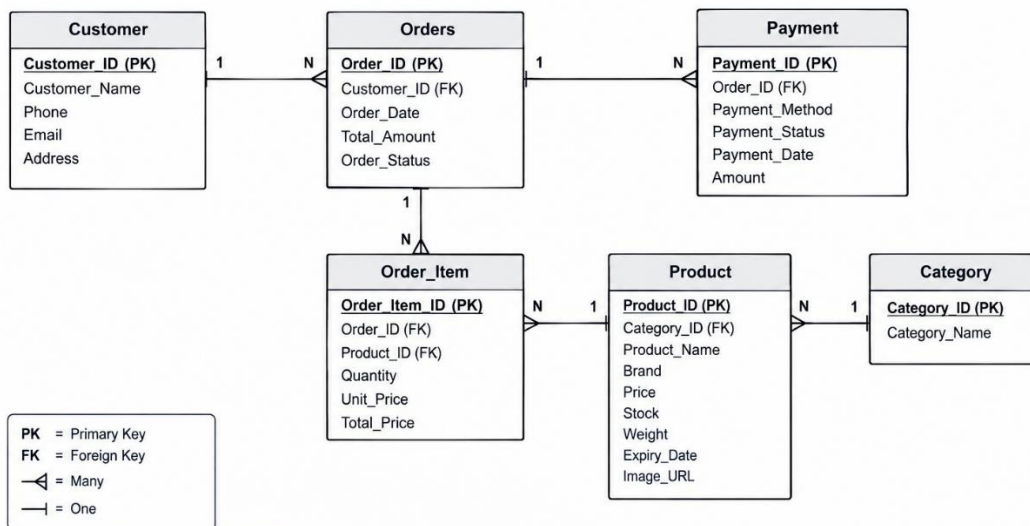
Swiggy Instamart E-Commerce Database Management System

1. Database Design

1.1 Payment Table

Attribute	Data Type	Key Type / Constraint
Payment_ID	NUMBER(10)	Primary Key
Order_ID	NUMBER(10)	Foreign Key
Payment_Method	VARCHAR2(30)	NOT NULL
Payment_Status	VARCHAR2(20)	NOT NULL
Payment_Date	DATE	—
Amount	NUMBER(10,2)	NOT NULL

2. Table Relationships



Relationships:

1. One Customer → Many Orders (Customer_ID → Orders.Customer_ID)
2. One Order → Many Payments (Order_ID → Payment.Order_ID)
3. One Order → Many Order Items (Order_ID → Order_Item.Order_ID)
4. One Product → Many Order Items (Product_ID → Order_Item.Product_ID)
5. One Category → Many Products (Category_ID → Product.Category_ID)

3. Table Creation

```
SQL> CREATE TABLE Payment (  
 2     Payment_ID NUMBER(10) PRIMARY KEY,  
 3     Order_ID NUMBER(10),  
 4     Payment_Method VARCHAR2(30) NOT NULL,  
 5     Payment_Status VARCHAR2(20) NOT NULL,  
 6     Payment_Date DATE,  
 7     Amount NUMBER(10,2) NOT NULL  
 8 );
```

Table created.

4. Payment Table Insertion

```
SQL> INSERT INTO Payment  
 2 VALUES (1, 1001, 'UPI', 'Paid', DATE '2026-09-15', 500);  
  
1 row created.  
  
SQL> INSERT INTO Payment  
 2 VALUES (2, 1002, 'Card', 'Paid', DATE '2026-09-16', 500);  
  
1 row created.  
  
SQL> INSERT INTO Payment  
 2 VALUES (3, 1003, 'Cash', 'Paid', DATE '2026-09-17', 320);  
  
1 row created.  
  
SQL> INSERT INTO Payment  
 2 VALUES (4, 1004, 'UPI', 'Paid', DATE '2026-09-18', 150);  
  
1 row created.  
  
SQL> INSERT INTO Payment  
 2 VALUES (5, 1005, 'Card', 'Paid', DATE '2026-09-19', 120);  
  
1 row created.  
  
SQL> INSERT INTO Payment  
 2 VALUES (6, 1006, 'UPI', 'Pending', DATE '2026-09-20', 80);  
  
1 row created.  
  
SQL> INSERT INTO Payment  
 2 VALUES (7, 1007, 'Cash', 'Paid', DATE '2026-09-21', 200);  
  
1 row created.  
  
SQL> INSERT INTO Payment  
 2 VALUES (8, 1008, 'UPI', 'Paid', DATE '2026-09-22', 320);  
  
1 row created.
```

5. Manage Payment Transactions

5.1 Display Successful Payments

```
SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_Status = 'Paid';
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS
1	1001	UPI	Paid
15-SEP-26	500		
2	1002	Card	Paid
16-SEP-26	500		
3	1003	Cash	Paid
17-SEP-26	320		

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS
4	1004	UPI	Paid
18-SEP-26	150		
5	1005	Card	Paid
19-SEP-26	120		
7	1007	Cash	Paid
21-SEP-26	200		

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS
8	1008	UPI	Paid
22-SEP-26	320		

7 rows selected.

6. Modify Payment Status

```
SQL> UPDATE Payment
  2 SET Payment_Status = 'Paid'
  3 WHERE Payment_ID = 6;
```

1 row updated.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> UPDATE Payment
  2 SET Payment_Status = 'Paid'
  3 WHERE Payment_ID = 10;
```

1 row updated.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_ID IN (6, 10);
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS
6	1006	UPI	Paid
20-SEP-26	80		
10	1002	UPI	Paid
24-SEP-26	500		

7. Analyze Payment Methods

```
SQL> SELECT Payment_Method,  
2          COUNT(*) AS Total_Payments  
3 FROM Payment  
4 GROUP BY Payment_Method  
5 ORDER BY Total_Payments DESC;
```

PAYMENT_METHOD	TOTAL_PAYMENTS
UPI	5
Card	3
Cash	2

Calculate Total Amount by Payment Method

```
SQL> SELECT  
2      p.Payment_Method,  
3      SUM(o.Total_Amount) AS Total_Amount  
4 FROM Payment p  
5 INNER JOIN Orders o  
6      ON p.Order_ID = o.Order_ID  
7 GROUP BY p.Payment_Method  
8 ORDER BY Total_Amount DESC;
```

PAYMENT_METHOD	TOTAL_AMOUNT
UPI	2980
Card	2270
Cash	1195

8. Payment Transaction Report

```

SQL> SELECT
2     p.Payment_ID,
3     p.Order_ID,
4     o.Order_Date,
5     o.Total_Amount,
6     p.Payment_Method,
7     p.Payment_Status,
8     p.Payment_Date
9  FROM Payment p
10 INNER JOIN Orders o
11     ON p.Order_ID = o.Order_ID
12 ORDER BY p.Payment_Date;

```

PAYMENT_ID	ORDER_ID	ORDER_DAT	TOTAL_AMOUNT	PAYMENT_METHOD
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PAYMENT_STATUS	PAYMENT_D
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Paid	1	1001 15-SEP-26 15-SEP-26	500	UPI
Paid	2	1002 16-SEP-26 16-SEP-26	850	Card
Paid	3	1003 17-SEP-26 17-SEP-26	320	Cash

PAYMENT_ID	ORDER_ID	ORDER_DAT	TOTAL_AMOUNT	PAYMENT_METHOD
------------	----------	-----------	--------------	----------------

PAYMENT_STATUS	PAYMENT_D
----------------	-----------

Paid	4	1004 18-SEP-26 18-SEP-26	650	UPI
Paid	5	1005 19-SEP-26 19-SEP-26	920	Card
Paid	6	1006 20-SEP-26 20-SEP-26	550	UPI

PAYMENT_ID	ORDER_ID	ORDER_DAT	TOTAL_AMOUNT	PAYMENT_METHOD
------------	----------	-----------	--------------	----------------

PAYMENT_STATUS	PAYMENT_D
----------------	-----------

Paid	4	1004 18-SEP-26 18-SEP-26	650	UPI
Paid	5	1005 19-SEP-26 19-SEP-26	920	Card
Paid	6	1006 20-SEP-26 20-SEP-26	550	UPI

PAYMENT_ID	ORDER_ID	ORDER_DAT	TOTAL_AMOUNT	PAYMENT_METHOD
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PAYMENT_STATUS	PAYMENT_D
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Paid	7	1007 21-SEP-26 21-SEP-26	875	Cash
Paid	8	1008 22-SEP-26 22-SEP-26	430	UPI
Refunded	9	1001 15-SEP-26 23-SEP-26	500	Card

PAYMENT_ID	ORDER_ID	ORDER_DAT	TOTAL_AMOUNT	PAYMENT_METHOD
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PAYMENT_STATUS	PAYMENT_D
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Paid	10	1002 16-SEP-26 24-SEP-26	850	UPI
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10 rows selected.